DAY 03: MEDIAN & MODE (THE ROBUST MEASURES)

Bilingual Deep-Dive Notes | Mission 2029

1. Why Median? (Median Kyun Zaroori Hai?)

In Day 02, we saw that 'Mean' is sensitive to Outliers. Median is the 'Robust' alternative that ignores extreme values.

Day 02 mein humne dekha ki 'Mean' outliers se darr jata hai. Median ek aisa mazboot vikalp hai jo extreme values (outliers) ko ignore kar deta hai.

Median is the exact middle value of a dataset when arranged in order.

Median kisi bhi data set ki wo value hai jo theek beech mein (middle) aati hai jab hum data ko chhote se bade kram mein lagate hain.

2. How to find Median? (Median Nikalne Ka Tarika)

Step 1: Sorting

Always arrange the data in Ascending Order (Smallest to Largest).

Hamesha data ko badhte kram (Ascending Order) mein lagayein.

Step 2: Position Check

Case A (Odd Number of Items): The middle one is the Median.

Case A (Visham/Odd Ginti): *Jo theek beech mein hai, wahi Median hai.*

Case B (Even Number of Items): Take the average of the two middle values.

Case B (Sam/Even Ginti): *Beech ki do values ka average (Mean) nikaalein.*

Jewellery Case Study: Median vs Mean

Weights of 5 rings: 4g, 5g, 5g, 6g, 80g (Outlier).

5 rings ka wazan: 4g, 5g, 5g, 6g, 80g (Outlier).

Mean: 20g (Misleading due to 80g).

Mean: *20g (80g ki wajah se galat tasveer).*

Median: Data sorted hai. Middle value 3rd number par hai = 5g.

Median: *Data sorted hai. Beech ki value (3rd) =* ***5g*** *(Ekdam Sahi!).*

3. What is Mode? (Mode Kya Hai?)

Mode is the value that appears most frequently in a dataset.

Mode wo value hai jo data mein sabse zyada baar (most frequent) aati hai.

It helps in identifying the 'Most Popular' or 'Common' item.

Ye sabse 'Popular' ya 'Common' item ko pehchanne mein madad karta hai.

Analyst Insight: Mode is the only measure that can be used for Qualitative (Categorical) data.

Analyst Insight: *Mode akela aisa measure hai jise hum Qualitative (Categorical) data ke liye bhi use kar sakte hain. Jaise: "Sabse zyada bikne wali metal?" (Gold).*

4. Comparison Table (Tulanatmak Adhyayan)

Measure	Best Used When... (Kab use karein?)	Impact of Outliers (Outliers ka asar)
Mean	Data is symmetric (Normal).	High (Bahut zyada)
Median	Data has outliers (Skewed).	No Impact (Koi asar nahi)
Mode	Dealing with Categories/Trends.	No Impact (Koi asar nahi)

5. Python Origin Logic (Preview)

To find Median in Python without libraries:

Bina library ke Python mein Median kaise nikaalein:

```
data.sort()
n = len(data)
# Find the middle index...
```

Day 03 Mission: By the end of today, you will know exactly which number represents your business truth.

Day 03 Mission: *Aaj ke din ke khatam hone tak, aapko pata hoga ki kaun sa number aapke business ki asli sachai dikhata hai.*